

Theme Discovery from Scratch

Motivation, taxonomy, and lifecycle

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1. Thematic trading

Why thematic trading?

Traditional investing organises companies along **time-invariant** axes: geography, sector, market cap, style (value/growth).

- Easy to benchmark, easy to index.
- Slow to capture *structural shifts* that cut across sectors.

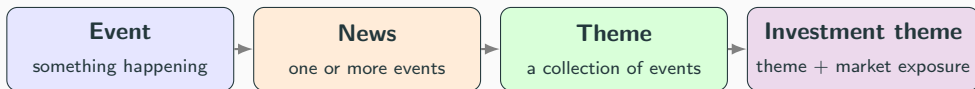
Thematic investing organises companies along a **narrative**: a structural change driving demand across multiple sectors at once.

- AI infrastructure $\Rightarrow$ chip designers + utilities + cooling + real estate
- Energy transition $\Rightarrow$ miners + grid operators + auto manufacturers
- GLP-1 obesity drugs $\Rightarrow$ pharma + food & beverage + medical devices

*The bet: catching the narrative **before** it is priced into every constituent.*

2. Taxonomy for theme detection

From news to theme — granularity ladder



- **News** is the raw input — articles bundle multiple events that contains one or more events.
- **Events** are single, datable happenings (a product launch, a policy announcement, an earnings beat).
- **Themes** are recurring patterns over events — a narrative binding them together.
- **Investment themes** add the second leg: a tradable basket whose value moves with the theme.

Sectors — the time-invariant baseline

Sectors classify *companies* along stable, slow-moving axes. Several competing taxonomies exist:

Standard	Maintained by	Granularity
GICS	MSCI / S&P	11 sectors → 25 industry groups → 74 industries
ICB	FTSE Russell	11 industries → 20 supersectors → 45 sectors
RBICS	FactSet	~1,500 leaf-level sub-industries (revenue-based)

Key property: a company's sector code rarely changes: **Time Invariant**.

What sectors cannot capture: cross-sector narratives, transient shifts, and companies whose thematic exposure is much larger than their sector classification suggests.

What is a theme? Two defining properties

A **theme** is a coherent narrative that ties together a stream of events. Two properties:

1. Transient (has a lifecycle)

Sectors are roughly permanent.

Themes emerge, accelerate, mature, and fade.

- *Cannabis* (2018–2021), now dormant
- *Metaverse* (2021–2022), faded fast
- *GLP-1* (2023–), still accelerating

2. Cross-sector

A theme groups companies that a sector taxonomy would scatter.

- *AI infrastructure*: semis (NVDA), utilities (VST), real estate (DLR), industrials (ETN)
- *Energy transition*: miners, grid operators, auto, chemicals

Operational definition for this project: a theme is a cluster of *events* that recur together over a sustained window and are mentioned in news with rising frequency.

Theme vs. investment theme

Theme

A narrative observable in the world.

- *The data exists*: news mentions, event clusters, co-occurrence patterns.
- Detectable from a news corpus alone.
- No requirement for any tradable instrument to exist.

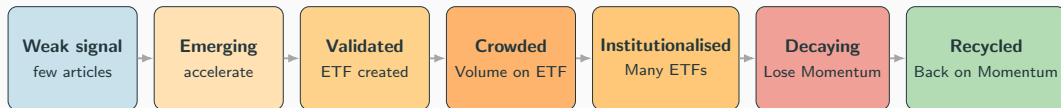
Investment theme

Theme + a defined market exposure.

- A *basket* (or ETF, or screen) whose value moves with the theme.
- Requires mapping the theme to companies (and weights).
- Trade-off: tighter basket $\Rightarrow$ more upside if right, but more risky.

Many themes never become investment themes (no clean basket exists, or the public-market exposure is too small). *Theme detection comes first; basket construction is a separate problem.*

The lifecycle of a theme



The tension: the earlier you detect, the more upside — but also the more noise. Most “weak signals” don’t become themes.

Where we want to operate: emerging → validated.

Clean Energy — full lifecycle (2018–2024)

Cross-sector by construction.

One narrative, beneficiaries in five different GICS sectors:

- **Utilities** — NextEra (NEE), Iberdrola (IBDRY)
- **Industrials** — Vestas (VWS), Enphase (ENPH)
- **Materials** — Albemarle (ALB), SQM (lithium)
- **Auto / consumer disc.** — Tesla (TSLA), BYD
- **Tech** — solar inverters, grid software

Lifecycle stage by stage.

- *Weak signal* 2018: niche ESG flows, falling solar cost
- *Emerging* 2019: Paris Agreement bites, EU Green Deal
- *Validated* 2020: Biden wins on climate platform
- *Crowded* early 2021: ICLN, TAN go vertical; HF flows
- *Institutionalised* 2021: every diversified book has ICLN
- *Decaying* 2022: rate hikes, supply-chain hits, peak euphoria fades

What you missed if you waited for the ETF stage.

ICLN (iShares clean-energy ETF): ~\$10 early 2020 → ~\$33 Jan 2021 (~3.3×).

TSLA: ~\$30 split-adj. early 2020 → ~\$400+ at peak.

ENPH: ~\$25 early 2020 → ~\$330 peak.

Most of the return happened in the *emerging* → *crowded* window (2019–early 2021). By Institutionalised (mid-2021), the trade was already over — ICLN halved through 2022.

The recycled wave (2023–24). Old constituents that decayed (TSLA, ENPH) didn't recover. But nuclear utilities (VST

AI — winning theme (2022–2024)

Cross-sector beneficiaries.

Obvious in semis — but the real edge came from the second-order plays:

- **Semis / hardware** — NVDA, AMD, TSM, AVGO
- **Software** — MSFT, GOOG, ORCL
- **Utilities / power** — VST, CEG, TLN (AI data-centre power demand)
- **Industrials** — VRT, ETN (cooling, electrical infra)
- **Real estate** — DLR, EQIX (data centres)
- **Materials** — copper miners (FCX) for grid build-out

Lifecycle stage by stage.

- *Weak signal* 2017–2019: Transformer paper, GPT-2; niche
- *Emerging* 2020–early 2022: GPT-3, scaling laws
- *Validated* Nov 2022: ChatGPT moment — narrative crystallises
- *Crowded* 2023: NVDA +200% over the HF long the

What catching the narrative early was worth.

NVDA: ~\$15 Oct 2022 → ~\$135 mid-2024 (~9×).

VST (utility, AI-power play): ~\$25 early 2023 → ~\$170 peak (~7×). Sector ETF XLU did ~+10% over the same window.

The *utilities* leg is the key insight: VST is in no AI-themed ETF at the start of 2023, invisible to a sector view. A cross-sector thematic lens was required to surface it.

Metaverse — counterexample (2021–2022)

Looked like a textbook cross-sector theme.

- **Tech** — Meta (FB→META), MSFT (HoloLens)
- **Gaming** — Roblox (RBLX), Unity (U)
- **Hardware** — Apple (Vision Pro), Sony
- **Consumer** — Nike (NFT push), luxury brands
- **Semis** — NVDA (Omniverse pitch)

Lifecycle that short-circuited.

- *Weak signal* 2014–2020: VR niche, Oculus acquired by Facebook
- *Emerging* Oct 2021: Facebook renames to Meta; Roundhill META ETF launches
- *Crowded* late 2021: peak hype, every CEO mentions it
- **Validated stage skipped**
- *Decaying* 2022: Reality Labs losses balloon
- *Recycled?* arguably as “spatial computing” / Apple Vision Pro — but mostly dormant

What “buying the theme” cost.

META (the stock): peak ~\$380 Sep 2021 → trough ~\$90 Nov 2022 (−75%).

RBLX: ~\$140 peak → ~\$25 trough.

META (the Roundhill ETF): launched near peak hype, never recovered.

Cross-sector *and* news velocity were both present. **What failed was validation**: the underlying demand never materialised. A pure news-acceleration signal would have flagged this — and been wrong.

Quantum Computing & diamonds — today's weak signal

The narrative.

Leading quantum-hardware architectures (NV-centre qubits, photonic quantum sensors) need **ultra-pure synthetic diamond substrates**. CVD-diamond producers become indirect quantum plays.

Potential cross-sector beneficiaries.

- **Tech** — IBM, IonQ, RGTI (the names everyone watches)
- **Materials / mining** — synthetic-diamond producers (mostly private; a few small caps)
- **Semis** — quantum-control electronics
- **Defense** — quantum sensing (navigation, ASW)

Lifecycle position today

Broad “Quantum” theme: *emerging* (1–2 ETFs launched, niche coverage).

“Diamond supply chain for quantum” sub-theme: *weak signal*.

A sector view would never surface mining companies as quantum plays.

The honest question. Both Clean Energy and Metaverse looked exactly like this at the weak-signal stage. Where do you want the precision/recall threshold of the system?

Discussion

Questions for you

- **Lifecycle target.** Where on the 7-stage cycle do you want signals to fire?
Weak/Emerging (high recall, noisy), Validated (balanced), Crowded (low noise but late), or Recycled (catching the second wave of a known theme)?
- **Theme universe.** Do you want a fixed list of themes to monitor, or open-ended discovery from the news corpus?
- **Sub-themes.** For mature themes like AI / Clean Energy, do you want the system to drill down into sub-themes automatically, or is that something your analysts do?
- **Cross-sector reach.** How important is it to surface cross-sector plays (like VST in utilities, or diamond miners for quantum) vs. sticking to the obvious constituents?
- **Output format.** Daily ranking, weekly digest, on-alert pings — what fits your workflow?